

25/75 YES/NO Stress Test Results on POPE Benchmark

Ground truth label ratio: 25% YES / 75% NO

Baseline is the reference point for FN→TP and TN→FP columns.

Column definitions:

- **FN→TP**: Change in correctly answered YES questions relative to baseline (positive = more correct YES answers recovered).
- **TN→FP**: Change in incorrectly answered YES questions relative to baseline (positive = more wrong YES answers introduced).
- **Yes%**: Proportion of questions answered “yes” by the model.
- **Acc%**: Overall accuracy on the 25/75 filtered benchmark.

Note: Each image retains 1 YES and 3 NO questions (2000 total per split). The same YES question ordinals are deleted consistently across all POPE variants (Random, Popular, Adversarial) and all decoding methods for each image (seed = 42). **CD methods:** VCD, SID, ICD. **Spurious baselines:** PBA (prompt: “Answer yes whenever possible.”), OLM (output label manipulation).

Table 1: 25/75 POPE stress test – **LLaVA-v1.5-7B** on **GQA**.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	26.2	91.85
	VCD	+23	+119	33.4	87.05
	SID	+19	+72	30.8	89.20
	ICD	−9	−16	25.0	92.20
	PBA	+16	+43	29.2	90.50
	OLM	+34	+137	34.8	86.70
Popular	Baseline	ref	ref	34.6	83.45
	VCD	+26	+135	42.7	78.00
	SID	+17	+61	38.6	81.25
	ICD	−8	−31	32.7	84.60
	PBA	+16	+72	39.1	80.65
	OLM	+34	+265	49.6	71.90
Adversarial	Baseline	ref	ref	39.6	78.55
	VCD	+21	+147	47.9	72.25
	SID	+17	+80	44.4	75.40
	ICD	−8	−36	37.4	79.95
	PBA	+16	+57	43.2	76.50
	OLM	+34	+287	55.6	65.90

Table 2: 25/75 POPE stress test – **LLaVA-v1.5-7B** on **COCO**. ICD outputs not available.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	20.6	91.60
	VCD	+41	+60	25.7	90.65
	SID	+21	+31	23.2	91.10
	PBA	+13	+3	21.4	92.10
	OLM	+48	+53	25.7	91.35
Popular	Baseline	ref	ref	22.2	89.95
	VCD	+45	+90	29.0	87.70
	SID	+24	+66	26.8	87.85
	PBA	+13	+22	24.0	89.50
	OLM	+48	+89	29.1	87.90
Adversarial	Baseline	ref	ref	26.0	86.20
	VCD	+41	+134	34.8	81.55
	SID	+27	+95	32.1	82.80
	PBA	+13	+26	28.0	85.55
	OLM	+47	+142	35.4	81.45

Table 3: 25/75 POPE stress test – **LLaVA-v1.5-7B** on **AOKVQA**.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	26.5	91.45
	VCD	+18	+107	32.7	87.00
	SID	+3	+68	30.0	88.20
	ICD	−9	−8	25.6	91.40
	PBA	+12	+24	28.2	90.85
	OLM	+30	+116	33.8	87.15
Popular	Baseline	ref	ref	31.7	86.20
	VCD	+15	+139	39.4	80.00
	SID	+4	+81	35.9	82.35
	ICD	−8	−24	30.1	87.00
	PBA	+12	+52	34.9	84.20
	OLM	+30	+204	43.4	77.50
Adversarial	Baseline	ref	ref	41.3	76.60
	VCD	+17	+146	49.5	70.15
	SID	+3	+81	45.5	72.70
	ICD	−10	−39	38.9	78.05
	PBA	+12	+66	45.2	73.90
	OLM	+30	+255	55.5	65.35

Table 4: 25/75 POPE stress test – **Qwen2.5-VL-7B** on **GQA**.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	24.3	94.15
	VCD	0	+8	24.8	93.75
	SID	+4	+16	25.4	93.55
	ICD	+2	+6	24.8	93.95
	PBA	+2	+2	24.6	94.15
	OLM	+24	+40	27.6	93.35
Popular	Baseline	ref	ref	30.2	88.35
	VCD	−3	+24	31.3	87.00
	SID	+3	+35	32.1	86.75
	ICD	0	+8	30.6	87.95
	PBA	+1	+7	30.6	88.05
	OLM	+23	+75	35.1	85.75
Adversarial	Baseline	ref	ref	34.5	84.05
	VCD	−2	+12	35.0	83.35
	SID	+3	+26	36.0	82.90
	ICD	0	+5	34.8	83.80
	PBA	+1	+7	34.9	83.75
	OLM	+23	+72	39.3	81.60

Table 5: 25/75 POPE stress test – **Qwen2.5-VL-7B** on **COCO**.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	20.1	93.55
	VCD	+10	+2	20.6	93.95
	SID	+9	+5	20.8	93.75
	ICD	−2	−1	19.9	93.50
	PBA	+5	0	20.3	93.80
	OLM	+29	+10	22.0	94.50
Popular	Baseline	ref	ref	21.6	92.05
	VCD	+8	+9	22.4	92.00
	SID	+9	+13	22.7	91.85
	ICD	−1	+4	21.7	91.80
	PBA	+5	0	21.8	92.30
	OLM	+29	+19	23.9	92.55
Adversarial	Baseline	ref	ref	23.0	90.60
	VCD	+7	+11	23.9	90.40
	SID	+9	+27	24.8	89.70
	ICD	−2	+3	23.1	90.35
	PBA	+5	+4	23.4	90.65
	OLM	+29	+44	26.7	89.85

Table 6: 25/75 POPE stress test – **Qwen2.5-VL-7B** on **AOKVQA**.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	24.9	94.25
	VCD	+2	+10	25.6	93.85
	SID	+4	+15	25.9	93.70
	ICD	−13	+3	24.4	93.45
	PBA	0	+2	25.1	94.15
	OLM	+21	+31	27.6	93.75
Popular	Baseline	ref	ref	28.1	91.15
	VCD	+1	+15	28.8	90.45
	SID	+4	+21	29.3	90.30
	ICD	−12	+2	27.6	90.45
	PBA	0	+9	28.5	90.70
	OLM	+21	+42	31.2	90.10
Adversarial	Baseline	ref	ref	37.1	82.10
	VCD	0	−4	36.9	82.30
	SID	+4	+11	37.9	81.75
	ICD	−10	0	36.6	81.60
	PBA	0	+9	37.5	81.65
	OLM	+21	+57	41.0	80.30